

Aleksei Gregory Sorokin

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Research Scientific Machine Learning, Gaussian Processes, Quasi-Monte Carlo, Probabilistic Numerics

Programming Python (PyTorch, GPyTorch, Pandas, Matplotlib), Julia, C, MATLAB, R, SQL, Wolfram

Tools AWS (SageMaker, EC2), GitHub (general, actions, pages), L^AT_EX, Docker

Education

2026.01 - 2028.05 **Postdoc in Statistics.** University of Chicago. Advisors *Yuehaw Khoo and Lek-Heng Lim*.

2021.08 - 2025.12 **PhD in Applied Math.** Illinois Institute of Technology (IIT). GPA 3.89/4. Advisor *Fred J Hickernell*.

2017.08 - 2021.05 **Master of Data Science.** IIT. Summa Cum Laude. GPA 3.94/4.

2017.08 - 2021.05 **B.S. in Applied Math, Minor in Computer Science.** IIT. Summa Cum Laude. GPA 3.94/4.

Experiences

2025.01 - 2025.12 **DOE SCGSR Fellow in Applied Math at Sandia National Laboratory** in Livermore, CA. I produced scientific ML models for machine-precision solutions to nonlinear PDEs [1]. I developed scalable multifidelity Gaussian processes regression models and open-source software implementations [8, 9, 5].

2024.05 - 2024.08 **Scientific Machine Learning Researcher at FM (Factory Mutual Insurance Company).** I deployed scientific ML models, including PINNs DeepONets, to accelerate CFD fire dynamics simulations [11].

2023.05 - 2023.08 **Graduate Intern at Los Alamos National Laboratory.** I modeled multifidelity solutions to PDE with random coefficients using efficient and error aware Gaussian processes regression models [12].

2022.05 - 2022.08 **Givens Associate Intern at Argonne National Laboratory.** I derived error bounds and a sequential sampling method for efficiently estimating failure probabilities with probabilistic models [4].

2021.05 - 2021.08 **ML Engineer Intern at SigOpt, an Intel Company.** In a six-person ML team, I contributed production code for meta-learning model-aware hyperparameter tuning via Bayesian optimization [13].

2022.09 - 2022.11 **Participant in Argonne National Laboratory's Course on AI Driven Science on Supercomputers.** Key topics included handling large scale data pipelines and parallel training for neural networks.

2018.05 - 2019.08 **Instructor for the STARS Computing Corps' Computer Discover Program.** I taught and developed curriculum for middle school and high school girls to learn programmatic thinking in Python.

2021.08 - 2025.01 **Teaching Assistant at IIT.** I led reviews for PhD qualifying exams in analysis and computational math.

Open-Source Software

QMCPy Quasi-Monte Carlo Python Software (qmcsoftware.github.io/QMCSoftware). I led dozens of col-laborators across academia and industry to develop QMC sequence generators, automatic variable transformations, adaptive error estimators, and diverse use cases [3, 6, 7, 14, 15, 16, 17, 10, 2].

FastGPs Scalable Gaussian Processes in Python (alegresor.github.io/fastgps). This supports GPU scaling, batched inference, hyperparameter optimization, multifidelity GPs, and efficient Bayesian cubature. FastGPs is the first package to implement GPs which require only $\mathcal{O}(n)$ storage and $\mathcal{O}(n \log n)$ computations compared to the typical $\mathcal{O}(n^2)$ storage and $\mathcal{O}(n^3)$ computations requirements [9, 8, 5].

QMCGenerators.jl Randomized Quasi-Monte Carlo Sequences in Julia (alegresor.github.io/QMCGenerators.jl).

QMCToolsCL Randomized Quasi-Monte Carlo Sequences in C/OpenCL (qmcsoftware.github.io/QMCToolsCL/).

TorchOrthoPolys Orthogonal Polynomials in PyTorch (alegresor.github.io/TorchOrthoPolys/) with GPU support.

Awards

2025.01 - 2025.12 **DOE SCGSR Fellow in Applied Math**, Sandia National Laboratory, Livermore California.

2025.01 **Karl Menger Student Award for Exceptional Scholarship (Graduate)**, IIT.

2024.01 **College of Computing Excellence in Dissertation Research**, IIT.

2024 **Teaching Assistant Award**, IIT.

2023.08 **Outstanding Math Poster**, Los Alamos National Laboratory.

2017.08 - 2025.05 **Deans List Member**, IIT, every semester.

References

PhD Advisor **Fred J. Hickernell** (hickernell@iit.edu) Vice Provost for Research and Professor of Applied Math, IIT.

Mentor **Nicolas W. Hengartner** (nickh@lanl.gov) Senior Scientist, Los Alamos National Lab.

Mentor **Michael J. McCourt** (mikemccourt1234@gmail.com) CTO and Co-Founder at Distributional.

Mentor **Pieterjan M. Robbe** (pmrobbe@sandia.gov) Senior Member of Technical Staff, Sandia National Lab.

- [1] Aras Bacho, Aleksei G. Sorokin, Xianjin Yang, Théo Bourdais, Edoardo Calvello, Matthieu Darcy, Alexander W. Hsu, Bamdad Hosseini, and Houman Owhadi. "Operator learning at machine precision". In: *Journal of Computational Physics* 566 (2026), p. 115240. ISSN: 0021-9991. DOI: 10.1016/j.jcp.2026.115240. URL: <https://www.sciencedirect.com/science/article/pii/S0021999126005917>.
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- [4] Aleksei G. Sorokin and Vishwas Rao. "Credible intervals for probability of failure with Gaussian processes". In: (2026). DOI: 10.48550/arXiv.2311.07733.
- [5] Aleksei G. Sorokin, Pieterjan Robbe, and Fred J. Hickernell. *Fast multitask Gaussian process regression*. 2026. DOI: 10.48550/arXiv.2603.16014.
- [6] Aleksei G. Sorokin. "Algorithms and scientific software for quasi-Monte Carlo, fast Gaussian process regression, and scientific machine learning". PhD thesis. Illinois Institute of Technology, 2025. DOI: 10.48550/arXiv.2511.21915.
- [7] Aleksei G. Sorokin. "QMCPy: a Python package for randomized low-discrepancy sequences, quasi-Monte Carlo, and fast kernel methods". In: (2025). DOI: 10.48550/arXiv.2502.14256.
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- [9] Aleksei G. Sorokin, Pieterjan Robbe, and Fred J. Hickernell. "Fast Gaussian process regression for high dimensional functions with derivative information". In: *Proceedings of the First International Conference on Probabilistic Numerics*. Ed. by Motonobu Kanagawa, Jon Cockayne, Alexandra Gessner, and Philipp Hennig. Vol. 271. Proceedings of Machine Learning Research. PMLR, 2025, pp. 35–49. URL: <https://proceedings.mlr.press/v271/sorokin25a.html>.
- [10] Fred J. Hickernell, Nathan Kirk, and Aleksei G. Sorokin. "Quasi-Monte Carlo methods: what, why, and how?" In: (2024). Ed. by Ben Feng Christiane Lemieux, pp. 3–40. DOI: 10.1007/978-3-032-10590-5_1.
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